

INCIDENT REPORT

IR-001 - Windows Brute Force Failed Logins

Enterprise SOC Lab - Wazuh SIEM/XDR with MITRE ATT&CK; Detection

Field	Value
Prepared by	Jagriti Banerjee
Report ID	IR-001
GitHub-ready file name	IR-001-Windows-Brute-Force-Failed-Logins.pdf
Classification	Public portfolio lab evidence
Incident status	Resolved - lab simulation
Severity	High
Primary tactic	Credential Access
Primary technique	T1110.001 - Brute Force: Password Guessing

This public version intentionally uses relative sequencing and GitHub-ready artifact names. Absolute lab date values have been excluded from the written report to keep the document clean and reusable for portfolio publishing.

Executive Summary

Wazuh SIEM/XDR detected repeated Windows authentication failures against the Windows endpoint in the Enterprise SOC lab. The failed logins were generated from the Ubuntu endpoint using an SMB authentication test loop. The activity produced Windows logon failure telemetry, which was collected by the Wazuh Windows agent, processed by the Wazuh Manager, and surfaced in the Wazuh Dashboard as a brute-force style authentication alert.

The event was assessed as a true positive lab simulation. Although no production asset was impacted, the behavior represents a realistic credential access pattern that could indicate password guessing, credential stuffing, or unauthorized access attempts in an enterprise environment.

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1. Incident Snapshot

Category	Details
Incident name	Windows Brute Force Failed Logins
Report ID	IR-001
Source host	Ubuntu endpoint - 192.168.56.103
Target host	Windows endpoint - 192.168.56.101
SIEM/XDR	Wazuh 4.7.5 all-in-one deployment
Detection status	Detected and triaged
Verdict	True positive - controlled lab simulation
Business impact	No production impact; educational lab only
Primary evidence	Wazuh Security Events, SMB failed-login terminal output, MITRE ATT&CK dashboard

The attack simulation was designed to create authentication failure telemetry, not to compromise a real system. The objective was to validate whether Wazuh could collect, normalize, alert, and support analyst investigation of repeated failed logins.

2. Lab Environment and Scope

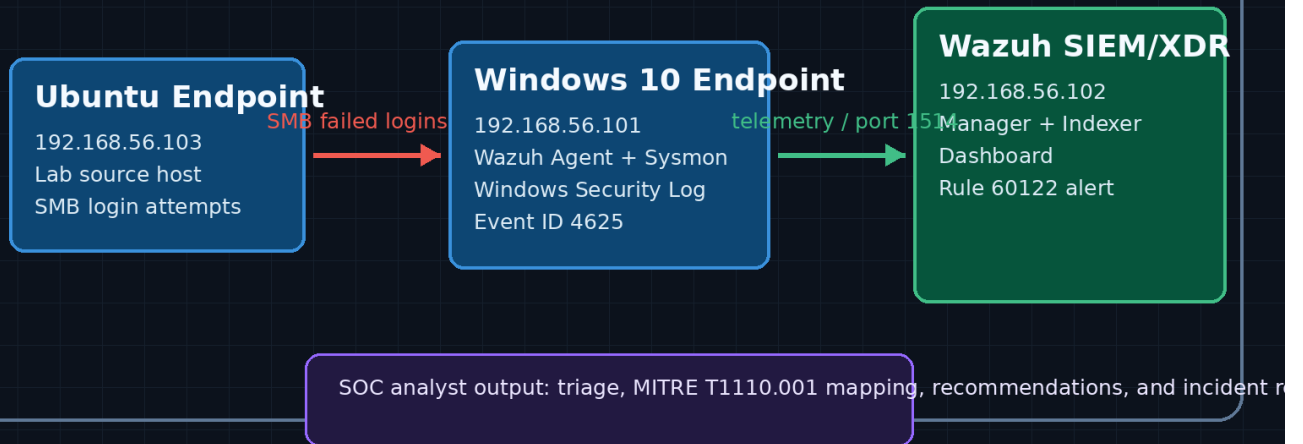
The environment was isolated inside a VirtualBox host-only lab network. All systems were controlled lab machines. No public target or third-party system was used.

Component	Role	IP / Identifier
Wazuh Server	Manager, Indexer, Dashboard, detection rules, alerting	192.168.56.102
Windows 10 Endpoint	Target endpoint receiving failed SMB authentication attempts	192.168.56.101
Ubuntu Endpoint	Lab source host used to generate failed SMB authentication attempts	192.168.56.103
Wazuh Agent	Forwards endpoint logs to Wazuh Manager	Agent transport on port 1514

IR-001 Detection Architecture - Windows Brute Force

Current lab flow: Ubuntu source host -> Windows endpoint -> Wazuh agent -> Wazuh SIEM/XDR -> SOC analyst

VirtualBox Host-only Lab Network - 192.168.56.0/24



Public/GitHub-ready version: absolute timestamps excluded; evidence names standardized.

Figure 1 - ir001-architecture-overview.png - GitHub-ready architecture evidence for IR-001.

3. GitHub-Ready Evidence Naming

The screenshot references below use lowercase, hyphen-separated names so that they can be placed directly inside a GitHub repository without spaces, brackets, apostrophes, or special characters. This prevents broken Markdown image links and makes the evidence folder easier to maintain.

Artifact purpose	GitHub-ready file name	Recommended repo path
Architecture overview	ir001-architecture-overview.png	screenshots/ir001-architecture-overview.png
Agents connected	ir001-agents-connected.png	screenshots/ir001-agents-connected.png
SMB trigger proof	ir001-smb-failed-login-trigger.png	screenshots/ir001-smb-failed-login-trigger.png
Detection pipeline	ir001-detection-pipeline.png	screenshots/ir001-detection-pipeline.png
Wazuh alert evidence	ir001-wazuh-brute-force-alert.png	screenshots/ir001-wazuh-brute-force-alert.png
MITRE mapping evidence	ir001-mitre-t1110-001-brute-force.png	screenshots/ir001-mitre-t1110-001-brute-force.png
Final report PDF	IR-001-Windows-Brute-Force-Failed-Logins.pdf	Reports/IR-001-Windows-Brute-Force-Failed-Logins.pdf

Naming standard used: lowercase words separated by hyphens, no spaces, no parentheses, no punctuation except hyphens, and file extension included.

4. Attack Narrative

The Ubuntu endpoint was used as a controlled lab source host to repeatedly attempt SMB authentication against the Windows endpoint. The attempted credentials failed, creating Windows logon failure activity. The Wazuh Windows agent collected the relevant security telemetry and forwarded it to the Wazuh Manager. Wazuh then displayed authentication failure alerts in the Security Events module.

This behavior simulates a common credential access pattern. In a real enterprise, repeated failed logins against a workstation or server may indicate password guessing, user enumeration, password spraying, or misconfigured automation.

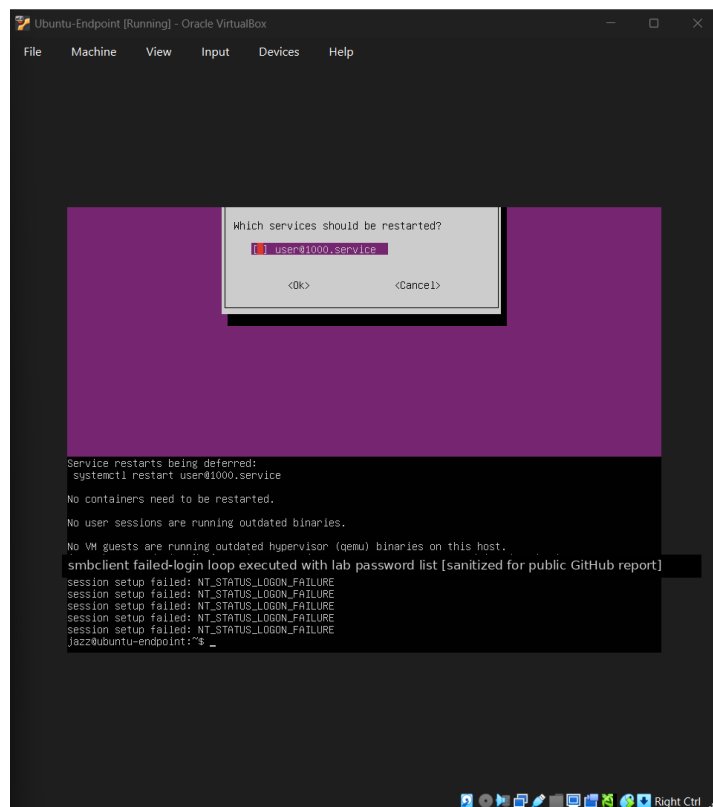


Figure 2 - ir001-smb-failed-login-trigger.png - Lab source host generated repeated failed SMB authentication attempts. Test password values are sanitized for public release.

5. Detection Pipeline

The detection pipeline connects the lab action to the final SOC output. The important concept is that Wazuh does not detect the attack by watching the screen. It detects log evidence generated by the endpoint and forwarded by the Wazuh agent.

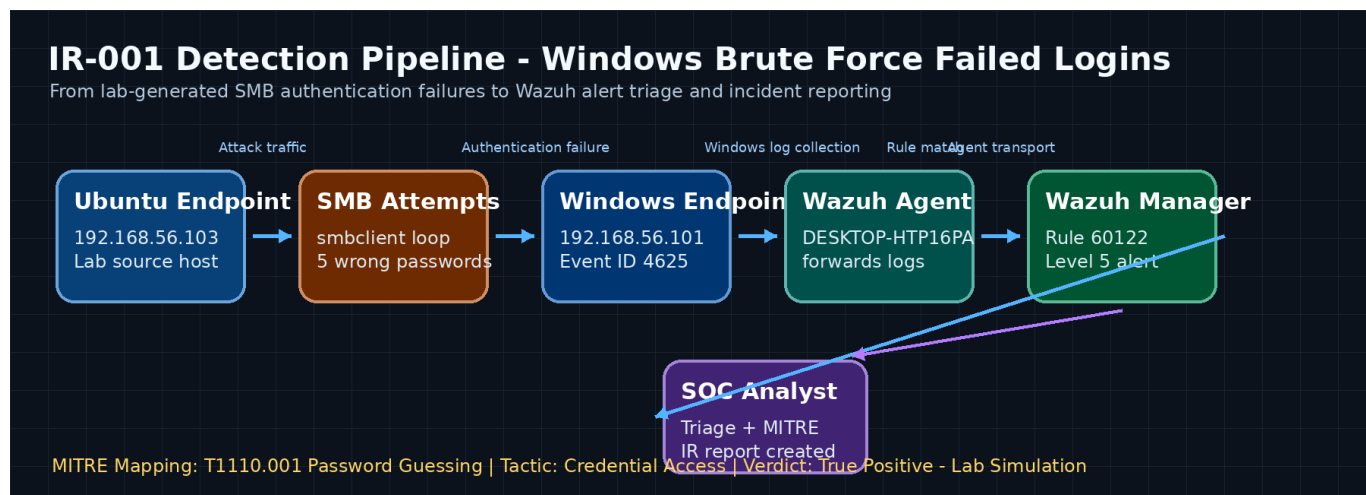


Figure 3 - ir001-detection-pipeline.png - IR-001 detection flow from failed SMB authentication to Wazuh alert and SOC reporting.

Stage	What happened	Why it matters
1. Trigger	Ubuntu generated repeated failed SMB authentication attempts.	Creates realistic credential access activity.
2. Endpoint log	Windows recorded failed logon telemetry.	Endpoint logs are the evidence source.
3. Agent	Wazuh agent forwarded events to the Wazuh Manager.	Centralizes endpoint telemetry.
4. Rule	Wazuh detection logic matched authentication failure behavior.	Turns raw logs into actionable alerts.
5. Triage	Alert was reviewed for host, rule, severity, MITRE mapping, and risk.	Converts alert into an incident decision.

6. Relative Timeline

The public report uses a relative sequence instead of absolute timestamps. This keeps the document clean and GitHub-ready while still showing the order of events.

Sequence	Event	Evidence
T+00	SMB failed-login simulation started from Ubuntu endpoint.	ir001-smb-failed-login-trigger.png
T+01	Windows endpoint recorded failed logon activity.	Windows Security Log / Wazuh Security Events
T+02	Wazuh agent forwarded Windows events to Wazuh Manager.	Agent transport / port 1514
T+03	Wazuh Security Events displayed repeated logon failure alerts.	ir001-wazuh-brute-force-alert.png
T+04	MITRE view mapped the behavior to brute force / password guessing.	ir001-mitre-t1110-001-brute-force.png
T+05	Analyst completed triage and documented incident report.	IR-001 report

7. Wazuh Alert Evidence

The Wazuh dashboard showed authentication failure alerts for the Windows endpoint. The key visible fields were the agent name, rule description, rule level, and rule ID. Absolute date fields are removed in the public evidence version.



Figure 4 - ir001-wazuh-brute-force-alert.png - Wazuh Security Events showing repeated Windows logon failures. Absolute date fields are removed for the public GitHub version.

Field	Observed / Interpreted Value
Agent	Windows endpoint - DESKTOP-HTP16PA
Target IP	192.168.56.101
Rule description	Logon failure - Unknown user or bad password
Observed rule ID	60122
Observed rule level	5
Detection meaning	Repeated failed Windows authentication events
Analyst interpretation	Credential access attempt / password guessing pattern in lab context

8. Agent Health Evidence

Before validating an alert, a SOC analyst must confirm that agents are connected and reporting. This prevents false assumptions caused by disconnected endpoints or missing telemetry.

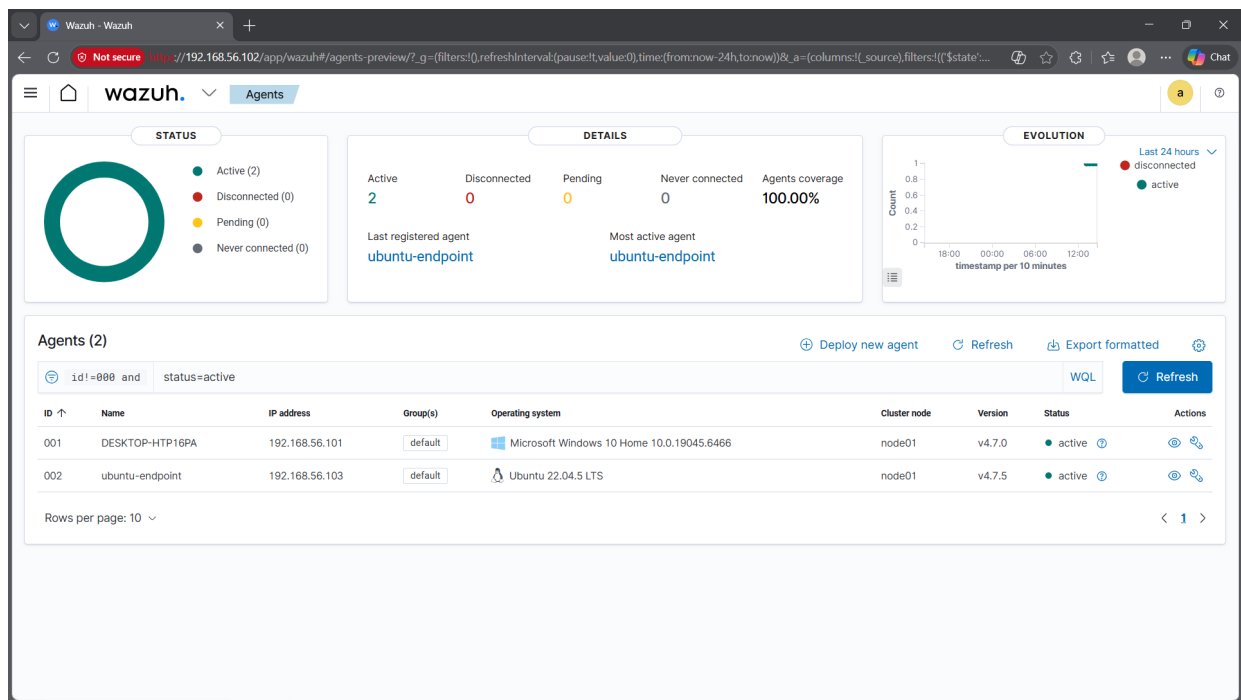


Figure 5 - ir001-agents-connected.png - Wazuh dashboard showing active Windows and Ubuntu agents.

9. MITRE ATT&CK; Mapping

The observed behavior maps to MITRE ATT&CK; Credential Access because the activity attempted to guess or validate credentials through repeated authentication attempts.

Field	Mapping
Tactic	Credential Access
Technique	T1110 - Brute Force
Sub-technique	T1110.001 - Password Guessing
Reason	Multiple failed authentication attempts were generated against a Windows endpoint.
Analyst confidence	High - lab-generated activity directly matches password guessing behavior.

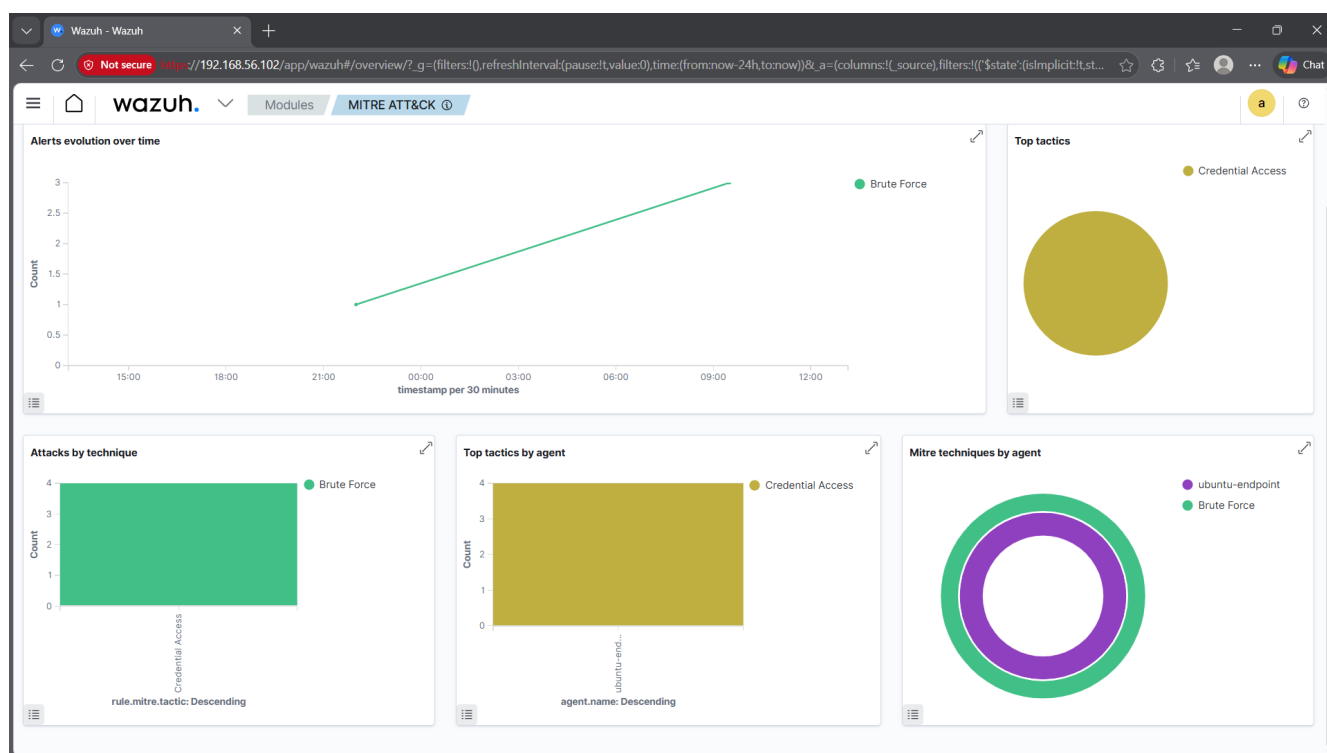


Figure 6 - ir001-mitre-t1110-001-brute-force.png - MITRE ATT&CK; dashboard showing brute force / credential access mapping.

10. Technical Analysis

Windows authentication failures are valuable security signals because they show authentication attempts that did not succeed. When these failures cluster around the same target, account, source host, or short time window, they may indicate password guessing or password spraying.

Technical item	Explanation
SMB authentication	The simulation used SMB authentication attempts from Ubuntu toward the Windows endpoint.
Windows failed logon	The Windows endpoint produced failed authentication telemetry that Wazuh could ingest.
Wazuh agent role	The agent forwarded the endpoint events to the Wazuh Manager.
Wazuh Manager role	The manager parsed events and applied detection rules.
Dashboard role	The dashboard presented alerts for analyst review and evidence capture.

Technical item	Explanation
MITRE value	MITRE mapping translated the alert into attacker behavior: Credential Access.

11. SOC Triage Notes

The alert was investigated using a standard SOC triage workflow. This section shows how a junior analyst should think through the alert rather than simply taking a screenshot.

Question	Answer for IR-001
What is the alert?	Repeated Windows logon failure events surfaced in Wazuh Security Events.
Which host was targeted?	Windows endpoint 192.168.56.101.
Which system generated the activity?	Ubuntu endpoint 192.168.56.103 in the lab.
Which rule / evidence supports it?	Wazuh Security Events show repeated logon failure entries with rule ID 60122.
What MITRE behavior is represented?	T1110.001 - Brute Force: Password Guessing.
Is it expected?	Yes, because this was intentionally triggered inside the lab.
Verdict?	True positive lab simulation.
Immediate response in production?	Review source, determine legitimacy, check for successful login, consider account lockout and blocking source.

12. Risk Assessment

Risk dimension	Assessment
Likelihood in real environment	Medium to High - brute-force attempts are common against exposed or reachable services.
Impact if successful	High - successful credential compromise could allow interactive access, lateral movement, data access, or persistence.
Observed lab impact	Low - controlled lab activity with no production exposure.
Security concern	Repeated failed logins may indicate an attacker attempting to discover or validate valid credentials.
Overall severity	High for behavior; contained in lab environment.

Severity rationale: The Wazuh alert level observed in the dashboard was lower than the business severity assigned here. A SOC report can escalate business severity when the behavior represents a high-risk attack pattern, even if a specific rule level is moderate.

13. Response Actions

Action	Lab status / production equivalent
Validate alert	Confirmed Wazuh displayed repeated Windows logon failures.
Confirm agent visibility	Windows and Ubuntu agents were active in Wazuh.
Identify source and target	Source: Ubuntu endpoint; Target: Windows endpoint.
Check for successful login	No confirmed successful attacker login was documented for this incident.
Document MITRE mapping	Mapped to Credential Access / T1110.001.
Report and close	Closed as resolved lab simulation with recommendations.

14. Recommendations

Priority	Recommendation	Reason
P1	Enable account lockout policy.	Limits repeated password guessing attempts.
P1	Monitor for successful login after repeated failures.	A successful login after failures may indicate credential compromise.
P1	Restrict SMB access to trusted hosts only.	Reduces attack surface.
P2	Enable MFA where possible.	Reduces impact of stolen or guessed passwords.
P2	Create Wazuh correlation rules for repeated failures from same source.	Improves detection accuracy and severity.
P2	Add Active Response for lab enrichment.	Can auto-block repeat brute-force sources in future lab phases.
P3	Export raw JSON evidence for each major alert.	Strengthens auditability and portfolio evidence.

15. GitHub Publishing Checklist

Item	Status guidance
PDF file name	Use Reports/IR-001-Windows-Brute-Force-Failed-Logins.pdf
Screenshot names	Use lowercase hyphenated names with .png extension.
Sensitive values	Do not expose real passwords, tokens, or personal credentials.
Time values	Use relative sequence in report; avoid unnecessary absolute date values in public text.
README link	Link report from Incident Reports table.
Evidence consistency	Ensure screenshot names in PDF match GitHub screenshot file names.

16. Appendix A - Reproduction Summary

The following is a high-level lab-only summary. It is included to explain the evidence chain and should only be performed against systems owned and controlled by the lab builder.

Lab-only trigger concept:

1. Use a controlled source VM inside the host-only network.
2. Attempt SMB authentication against the Windows endpoint using invalid lab credentials.
3. Confirm Windows logon failure events appear in Wazuh Security Events.
4. Validate the MITRE ATT&CK mapping.
5. Write incident report and recommendations.

17. Appendix B - Analyst Verdict

Verdict: True positive - controlled lab simulation.

Reason: The observed failed authentication activity was intentionally generated in the lab and was successfully detected by Wazuh. The behavior is representative of real-world password guessing and should be treated as high risk in a production environment.

18. Appendix C - Final Deliverable Names

Deliverable	GitHub-ready name
Report PDF	IR-001-Windows-Brute-Force-Failed-Logins.pdf
Architecture screenshot	ir001-architecture-overview.png
Agents screenshot	ir001-agents-connected.png
Trigger screenshot	ir001-smb-failed-login-trigger.png
Detection pipeline screenshot	ir001-detection-pipeline.png
Wazuh alert screenshot	ir001-wazuh-brute-force-alert.png
MITRE screenshot	ir001-mitre-t1110-001-brute-force.png